

QUIZ 3

5 questions and 1 min for each one

A researcher needs to find a large multi-omics cohort specifically focused on cancer genomic and clinical data. According to the course material, which repository is the most appropriate choice?

A. GDC / TCGA

B. PRIDE

C. GTEx

D. MetaboLights

In the NCBI Gene Expression Omnibus (GEO) system, which accession prefix identifies a complete study that may span multiple platforms?

A. SRR

B. GSE

C. GSM

D. GPL

Why does the course material advise against using *TPM* (Transcripts Per Million) or *FPKM* (Fragments Per Kilobase Million) as input for tools like *DESeq2*?

- A. These values do not include metadata such as subject ID or tissue type.
- B. These values are already normalized for library size and do not represent raw integer counts.
- C. These values are typically on a $\log_{10}$ scale which creates negative numbers.
- D. These formats can only be used for microarray data and are incompatible with RNA-seq.

In the suggested reproducible project structure, what is the strict 'Rule' for the ``data_raw/`` folder?

- A. All files in this folder must be manually renamed to match the publication's table names.
- B. Files should be read-only and never edited or renamed.
- C. It must contain a copy of the *R* scripts used for cleaning.
- D. It should only contain the final normalized matrices for analysis.

When using an AI agent to assist with bioinformatics tasks, which of the following is considered a task that **MUST NOT** be delegated blindly?

A. Generating the initial template for a reproducible folder structure.

B. Writing basic *R* code to calculate library sizes.

C. Drafting a checklist for quality control diagnostics.

D. The final scientific judgment and interpretation of study limitations.